



TECHNICAL UNIVERSITY

OF CLUJ-NAPOCA, ROMANIA

FACULTY OF AUTOMATION AND COMPUTER SCIENCE

MULTI-CHANNEL BIOPOTENTIAL DAQ WITH ROS2, IIO BINDINGS

DIPLOMA THESIS

Author: **Ioan Dragomir**

Scientific supervisor: **Șl. Dr. Ing. Tassos Natsakis**

2024



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Multi-channel biopotential DAQ with ROS2, IIO bindings

1. **The problem:** O scurtă descriere a temei proiectului de diplomă
2. **Content:** Presentation page, Declaration of authenticity, Summary, Contents, Introduction, Bibliographic research, Analysis design and implementation, Results and validation, Conclusion, Bibliography, Annexes
3. **Place of documentation:** Technical University of Cluj-Napoca
4. **Advisors:**
5. **Thesis emission date:** 02.07.2024
6. **Thesis delivery date:** 04.07.2024

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Declaration of authenticity

I, the undersigned, **Ioan Dragomir**, identified by the CI/BI with series RZ, number 113316, CNP 5011113410095, author of the thesis:

Multi-channel biopotential DAQ with ROS2, IIO bindings

conceived for the final graduation examination

at the **Faculty of Automation and Computer Science**,

the Systems Engineering specialization in English,

of the Technical University of Cluj-Napoca,

2023-2024 September session,

declare on my own responsibility that this thesis is the result of my own effort and intellectual activity, based on my own research and on information obtained from the sources cited both in the text and in the bibliography.

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Date

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SUMMARY

of the Diploma Thesis:

Multi-channel biopotential DAQ with ROS2, IIO bindings

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1. Requirements: A 8/16 channel ECG and EMG DAQ with at least 1000Hz sample rate, and good signal characteristics (SNR > 20dB, high CMRR, PSRR). Both hardware, firmware and software components designed and implemented. Open source hardware and software.
2. Solutions: Hardware: MAX78000 microcontroller, AD4114 ADC, custom designed and built pre-amplifier and Driven Right Leg modules. Firmware implements the libIIO protocol, providing a serial iio context. Software provides lightweight bindings for libIIO, Python, and a ROS2 publisher node.
3. Obtained results: All signal quality objectives are checked, with room for improvement for the sample rate when using all 16 channels, and software capabilities.
4. Tests and verification: The hardware components were tested individually using an ADALM2000-based test bench. The system as a whole was tested in various measurement scenarios: single/multiple channel EMG/ECG. The software components' throughput was tested to keep up with ADC output.
5. Personal contribution: Analog signal chain design. Firmware development and low level optimization of third party libraries. Thin software wrappers for interfacing with the rest of the system.
6. Bibliographical sources: M. Barbero, R. Merletti, and A. Rainoldi, "Atlas of Muscle Innervation Zones"; W. J. Esposito, "The ASPEN platform : Tools for mixed signal electronics, digital signal processing, and biomedical electronics"

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